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Enhancement of low quality underwater image through integrated global and local contrast correction

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ABSTRACT

The attenuation of the light that travels through a water medium subjects underwater images to several problems. As a result of low contrast and color performance, images are unclear and lose important information. Therefore, the objects in these images can hardly be differentiated from the background. This study proposes a new method called dual-image Rayleigh-stretched contrast-limited adaptive histogram specification, which integrates global and local contrast correction. The aims of the proposed method are to increase image details and to improve the visibility of underwater images while enhancing image contrasts. The two main steps of the proposed method are contrast and color corrections; an underwater image undergoes the former before the latter. Global contrast correction generates dual-intensity images, which are then integrated to produce contrast-enhanced resultant images. Subsequently, such images are processed locally to enhance details. The color of the images is also corrected to improve saturation and brightness. Qualitative and quantitative results show that the contrast of the resultant image improves significantly. Moreover, image detail and color are adequately enhanced; thus, the proposed approach outperforms current state-of-the-art methods.

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1. Introduction

The physical properties of an underwater medium prevent the degradation of normal images taken in the air [1]. As light travels in water, light intensity is exponentially lost depending on the wavelength of the color spectrum. The attenuation of light limits the visibility distance to approximately 20 m in clear water and to 5 m or less in turbid water [2]. Light that travels in air is partially reflected back upon entering water; the direction and effect vary based on the structure of the water surface [3]. In addition, water motion produces waves that diffuse the light entering the water to create crinkle patterns [4]. Light is attenuated exponentially with distance and depth mainly as a result of absorption and scattering effects [5].

In summary, the low quality of underwater images is mainly caused by the following factors: low contrast, blurring, the diminished true color of objects, bright artifacts, floating particles, and nonuniform lighting. These factors lead to unbalanced illumination. Consequently, various underwater imaging techniques and methods have been introduced into the field of underwater image processing.

2. Related works and of state-of-the-art problems in techniques for underwater image contrast enhancement

Underwater image processing can be categorized into two procedures: (i) image restoration and (ii) image enhancement [1]. The former focuses on recovering a degraded image by constructing a model of the degradation in the original image formation. Image restoration methods are exhaustive and require a few model parameters that characterize water turbidity. The latter uses qualitative and subjective criteria to produce a visually pleasing image without depending on a physical model for image formation.

The initial values of the darkest and brightest points play an important role in image color. Shamsudin et al. [3] and Rizzi et al. [6] proved that a black point takes an initial brightness value of 5% and that a white point has a value of 95%. Furthermore, Shamsudin et al. [3] identified a considerable difference in the correction techniques of auto-enhanced techniques and depth at a significance level of 5%. Manually enhanced techniques and depth also vary remarkably at a significance level of 1%. The significance level of 5% is discussed in subsequent sections; this level is applied by the

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proposed dual-image Rayleigh-stretched contrast-limited adaptive histogram specification (DIRS-CLAHS) method in the processing step.

Eustice et al. [7] proposed an extension method for the MATLAB image processing toolbox. In the current study, however, we focus on the first extension, to which Eustice et al. applied contrast-limited adaptive histogram specification (CLAHS) as a preprocessing step. In their study, these researchers confirmed that the Rayleigh distribution is ideal for representing underwater images [7]. Nevertheless, CLAHS has a drawback in that underwater images cannot be enhanced automatically because a few parameters should be set based on the image characteristics. Even though CLAHS enhances the underwater image effectively, this method diminishes the color of the output image. Moreover, much noise is produced in this image.

Hitam et al. [8] proposed a method called mixture contrast-limited adaptive histogram equalization (CLAHE-Mix) to reduce the significant noise introduced into underwater images by CLAHE. CLAHE-Mix enhances the contrast in underwater images and limits the induced artifacts generated with CLAHE [8]. Hitam et al. [8] also applied CLAHE to images in red–green–blue (RGB) and hue–saturation–value (HSV) color models separately. Then, the individual images are integrated through Euclidean distance to produce low-noise, contrast-enhanced images. In several cases, however, this method produces output images with more noise than the conventional CLAHE does. As such, the output image is greenish.

Pixel distribution shifting color correction (PDSCC) was proposed for digital color images by Naim and Isa [9]. This method corrects the white reference point of images to ensure that this point is achromatic. In the experiment conducted by these researchers, the resultant images exhibited better contrast and brightness than those produced with the gray world and white patch methods did. Nonetheless, PDSCC does not significantly enhance image contrast, unlike methods such as CLAHS and CLAHE-Mix.

Iqbal et al. presented the integrated color model (ICM) [10] and the unsupervised color correction method (UCM) [11]. When ICM is used, the input image in the RGB color model is decomposed into its respective channels before these channels are stretched over the entire dynamic range. Then, the image is converted into the hue–saturation–intensity (HSI) color model, where the S and I components are applied with contrast stretching throughout the entire dynamic range. These researchers also modified two color channels, namely, red and green, to reduce the color cast based on the von Kries hypothesis [11]. The overall observation indicates that the images are under-enhanced in several areas. The primary drawback of these reviewed techniques is that they produce resultant images corrupted with high noise.

In addition, Rizzi et al. [12] proposed an unsupervised digital image color equalization method simultaneous global and local effects. Schechner and Karpel [13,14] developed an image recovery algorithm based on a couple of images taken through a polarizer at different orientations by analyzing the physical effect of visibility degradation. Furthermore, Trucco and Olmos-Antillon [15] devised a self-tuning image restoration filter that simplified the well-known underwater image formation model proposed by Jaffe [16] and McGlamery [17]. However, the main disadvantages of these physics-based methods are that they require much computing resources and have a long execution time.

On the basis of previous research, two main problems have been identified in relation to the current underwater image contrast enhancement technique. First, the contrast of the output image remains low in the under-enhanced and over-enhanced areas. Second, several of these methods still generate noise/unwanted artifacts in the resultant image. Computational consumption is still high; as a result, processing time increases. The method proposed in the present study reduces the first two problems to produce underwater images with high contrast and little noise effect.

Underwater images normally exhibit a high percentage of blue, followed by green and red. Therefore, most underwater images appear bluish or greenish, as shown in Figs. 1 and 2, given that blue and green are the dominant color channels forming the overall image color. Red is the inferior color channel, and its percentage is generally lower than those of the other two color channels. This phenomenon can be observed in Fig. 1(b) and (c), where the pixels of the red channel are distributed on the left side (low intensity value) of the intensity diagram. The pixels of the blue and green color channels are positioned at high intensity level (on the right side of the intensity diagram), as per Fig. 1(b), (d), and (e).

Another underwater image sample, branch, is presented in Fig. 2 to explain the phenomenon further. In Fig. 2(b), the 3D RGB color model of this image clearly indicates that the pixels are distributed around the blue-green plane. Thus, a solution for improving the contrast of this underwater image involves modifying pixel concentration to ensure that the pixels are significantly distributed to appropriate color channels and are not concentrated only in certain channels or planes.

On the basis of the previously presented argument and hypothesis, this study emphasizes the image contrast enhancement technique that modifies image histograms without forming physical images.

Several state-of-the-art methods have successfully reduced the problem of underwater images. The results obtained for the image *branch* (Fig. 3) are analyzed to determine the advantages and disadvantages of these methods. The CLAHE-Mix method does not enhance image contrast adequately because the blue-green illumination effect is retained in the resultant image. In addition, the objects in the image are hardly distinguishable from the background, and the appearance of the image details is thus diminished. PDSCC generates a reddish image because pixel distribution shifts toward the red plane. The resultant image fails to improve image contrast given that the objects cannot be observed clearly.

The UCM and ICM approaches increase image contrast more effectively than the CLAHE-Mix and PDSCC methods do. Objects are effectively differentiated from the background; nevertheless, the UCM produces excessive red color. Therefore, the resultant image is reddish and yellowish. In addition, the blue-green illumination effect is retained in the image produced through ICM. Both of these methods over-enhance areas because excessively bright regions are spotted in the resultant images.

CLAHS [18] is commonly integrated with other techniques, such as the CLAHE-Mix method. CLAHS is utilized because of its promising results in enhancing image contrasts locally; nonetheless, certain parameters in this method, such as clip limit, distribution parameter, and number of tiles, should be set accordingly to enhance the resultant image contrast. The sample image depicted in Fig. 3 applies the default parameter setting for CLAHS [7] and CLAHE-Mix [8] as suggested by the authors. The image processed by CLAHS (Fig. 3) indicates that CLAHS improves the limited image contrast but does not reduce the blue-green illumination effect adequately.

The technique proposed in the present study is specifically designed for low-contrast underwater images that are normally dominated by green and blue color channels. As per previous research and the results presented in Fig. 3, enhancing image contrast typically results in an over-enhanced image that is reddish or excessively bright and is subject to the blue-green illumination effect. Furthermore, the output images appear unnatural.

These problems with underwater images cannot be solved by using only one general method. As per the previous explanation

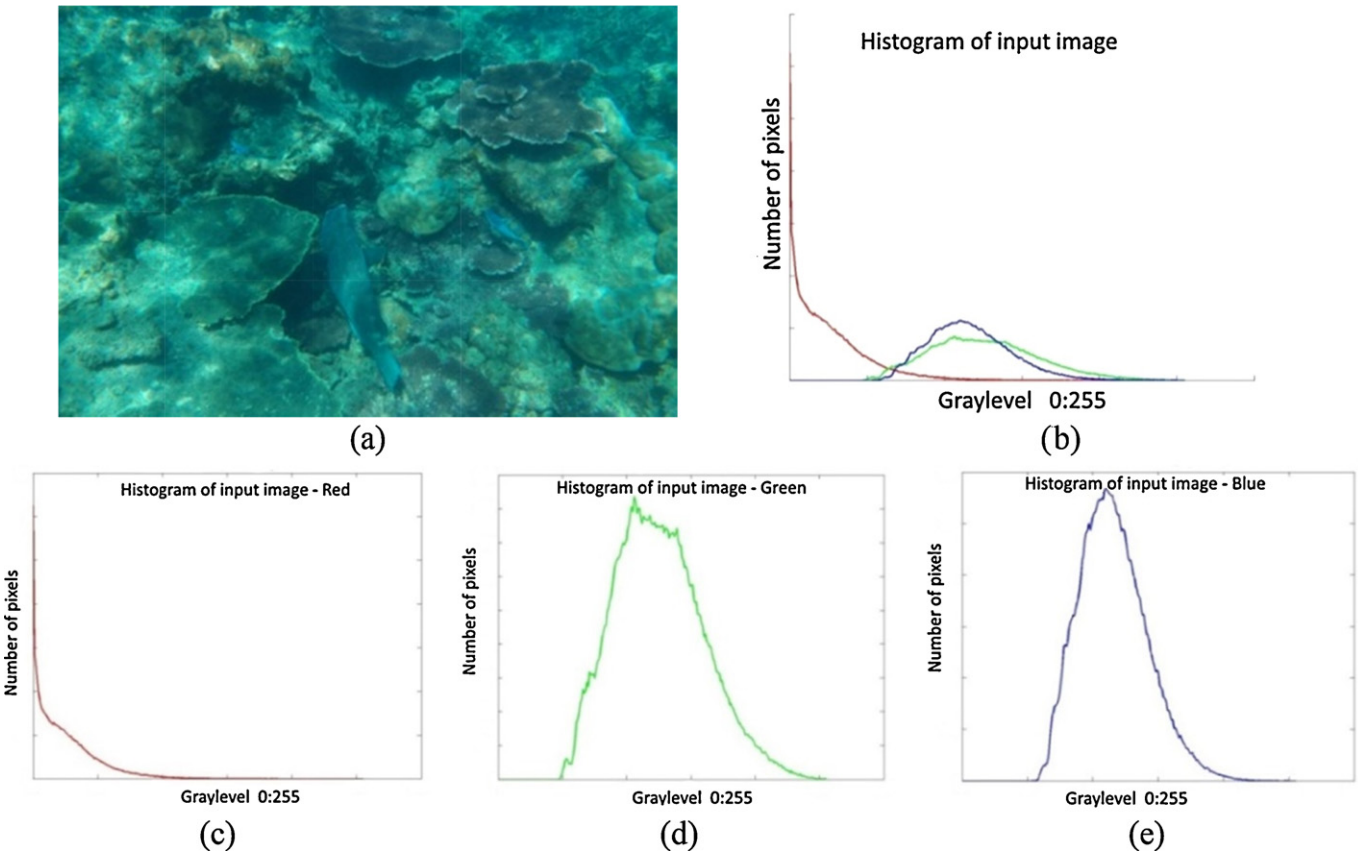

Fig. 1. Sample of underwater image *fish* with respective histogram of color channels (a) original image, (b) pixels distribution of RGB color channel pixels, and the distribution of (c) red, (d) green and (e) blue color channel pixels.

and the review presented in Section 2, the existing established methods have drawbacks although overall, these techniques can solve one or two specific underwater problems. By contrast, all the aforementioned problems are addressed by the proposed method, which reduces underwater image problems to a certain extent at minimum. Therefore, this approach is better than the current state-of-the-art techniques, as highlighted in the previous section and detailed in subsequent sections.

3. Motivation

The studies conducted by Iqbal et al. [10,11] and Abdul Ghani and Mat Isa [29] indicate a limitation in the stretching process, especially when the original histogram of the image is already stretched out to the entire dynamic range or almost the entire range. A widely stretched original histogram reduces the effectiveness of the stretching process because the histogram can be stretched only

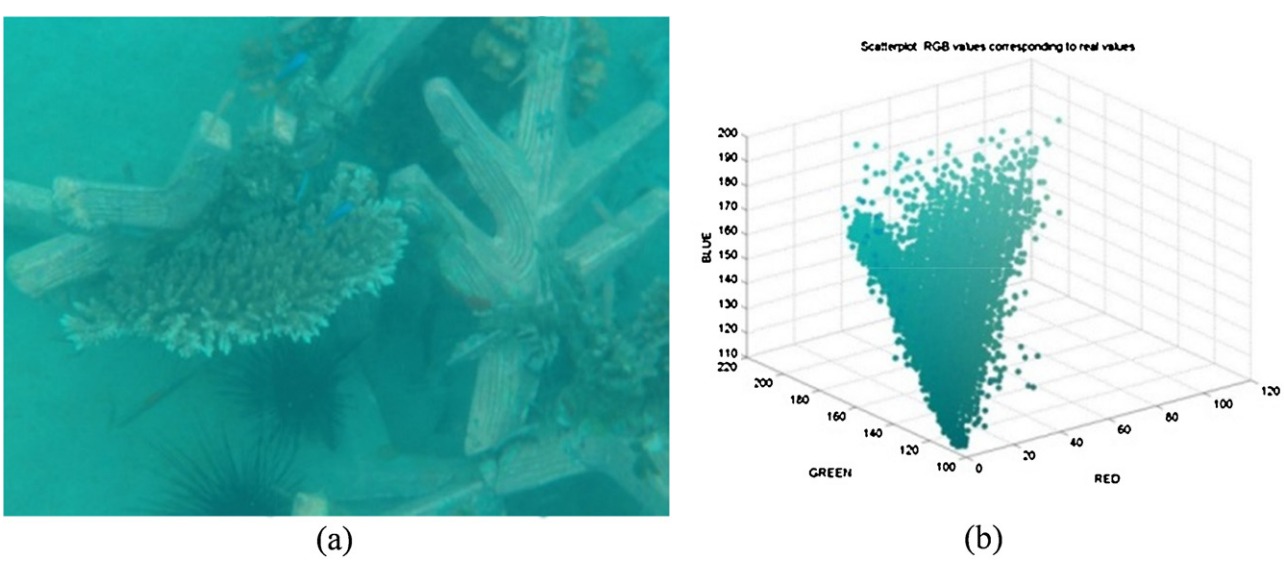

Fig. 2. Sample of an underwater image with a representation in the 3D RGB color model. (a) Original image *branch* and (b) representation of the image in the 3D RGB color model.

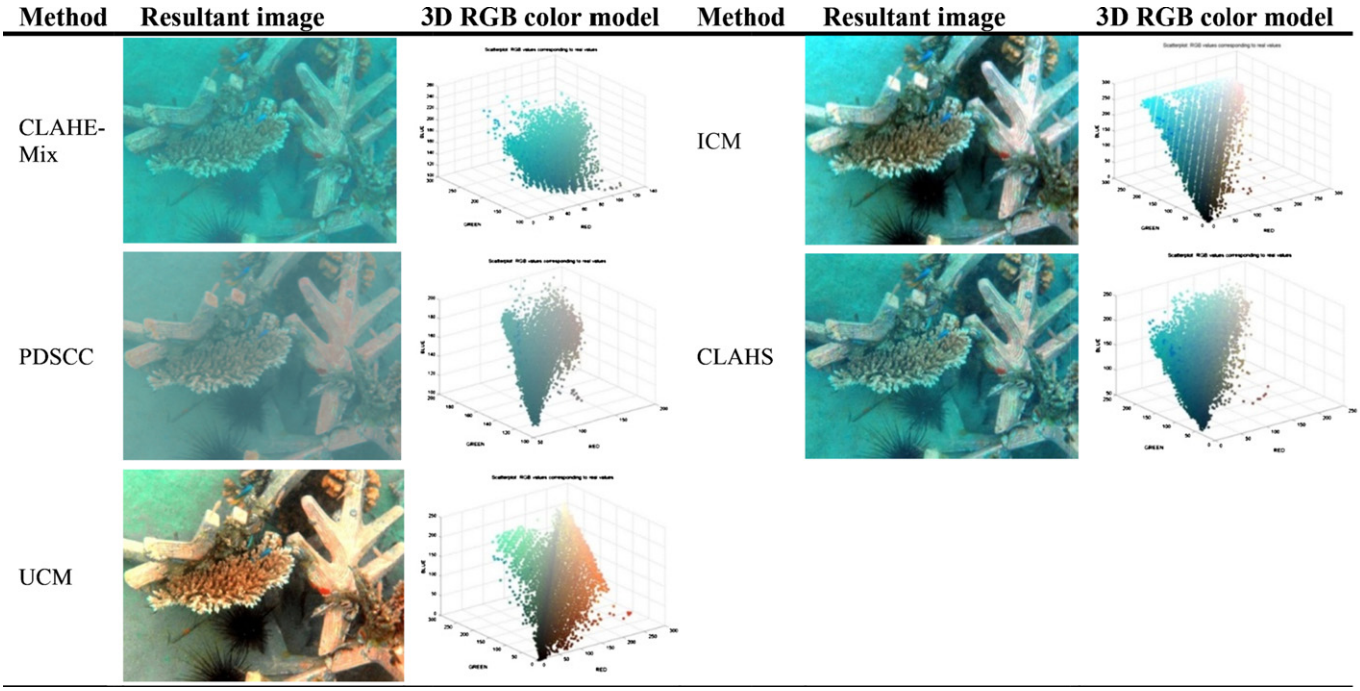

Fig. 3. Resultant *branch* images obtained by using current state-of-the-art techniques accompanied by their corresponding 3D RGB color models.

within a small range. In several cases, the pixels of the original histogram are mostly scattered on one side of the dynamic range; thus, pixel distribution is unbalanced given that the pixels are still concentrated although the conventional stretching process is applied. Therefore, pixel concentration can be reduced, along with the number of under-enhanced and over-enhanced areas.

Underwater images consist of under-enhanced and over-enhanced areas. Two images with such areas are to be produced from a single image. In addition, the histograms of underwater images are normally concentrated on a certain dynamic range based on their color channels, as indicated in Fig. 1. Thus, the influence of a color channel can be increased by stretching the histogram toward the high dynamic range. A similar process is performed with an upper-stretched histogram to reduce the domination of a certain color channel in a resultant image. The stretching process is conducted within the limit of 5%, according to the study conducted by Shamsudin et al. [3].

Under-enhancement and over-enhancement cannot be eliminated completely by global histogram stretching throughout the division and stretching processes. However, its effects can theoretically be reduced. To improve the enhancement process, the proposed method is integrated with local contrast correction, which is detailed in the subsequent section.

4. Integrated global and local contrast correction

4.1. Overview of the proposed technique

The study conducted by Abdul Ghani and Mat Isa [29] improves on the ICM [10] and UCM [11] methods previously proposed by Iqbal et al. The original ICM stretches underwater images in the RGB and HSI color models. By contrast, UCM stretches underwater images in the RGB and HSI color models within 0.1% of the input image after incorporating von Kries theory into the input image. The method proposed in [29] improves on ICM and UCM by implementing Rayleigh-stretched distribution within a certain range toward the output image histograms based on the results of [3].

In the current study, the proposed DIRS-CLAHS method is used to improve the enhancement technique of [29] by conducting global contrast correction at the beginning of the process to improve the overall image contrast. In addition to Rayleigh distribution, the integration of local contrast correction (CLAHS) into the proposed method is believed to enhance the local contrast of the image and to reduce image noise. The implementation of dual-intensity image composition in the HSV color model by the division and composition of the image saturation (S) and image brightness (V) components may improve image color, as described in this paper.

As displayed in Fig. 4, the DIRS-CLAHS method consists of two main stages, namely, (i) contrast correction and (ii) color correction. In the contrast correction stage, DIRS-CLAHS applies the global and local contrast correction methods. Global contrast stretching is employed in a manner that varies from the conventional process. The original histogram is divided at its midpoint to generate two different regions, namely, the lower and upper regions. In addition to the stretching process, the histogram regions are stretched to a limit of 5% from the minimum or maximum point. The lower regions are stretched toward the upper direction of the intensity level with a minimum value of 5% from the lowest intensity value (i.e., 5% from 0). By contrast, the upper regions are stretched toward the lower direction of the intensity level with a maximum value of 5% from the highest intensity value (i.e., from 255 for an 8 bit image). These limits are set to weaken the influence of dominant color channels and to enhance the inferior color channel. The image is processed further through local contrast correction. The CLAHS technique is applied to the image.

Then, the image is processed with the color correction method. The image is converted into the HSV color model, in which S and V can be modified directly to address under-enhancement and over-enhancement. These components are stretched to produce two different histograms with varying levels of saturation or brightness. Then, these histograms are composed to generate an output histogram by using the average values. The process is detailed in the subsequent section.

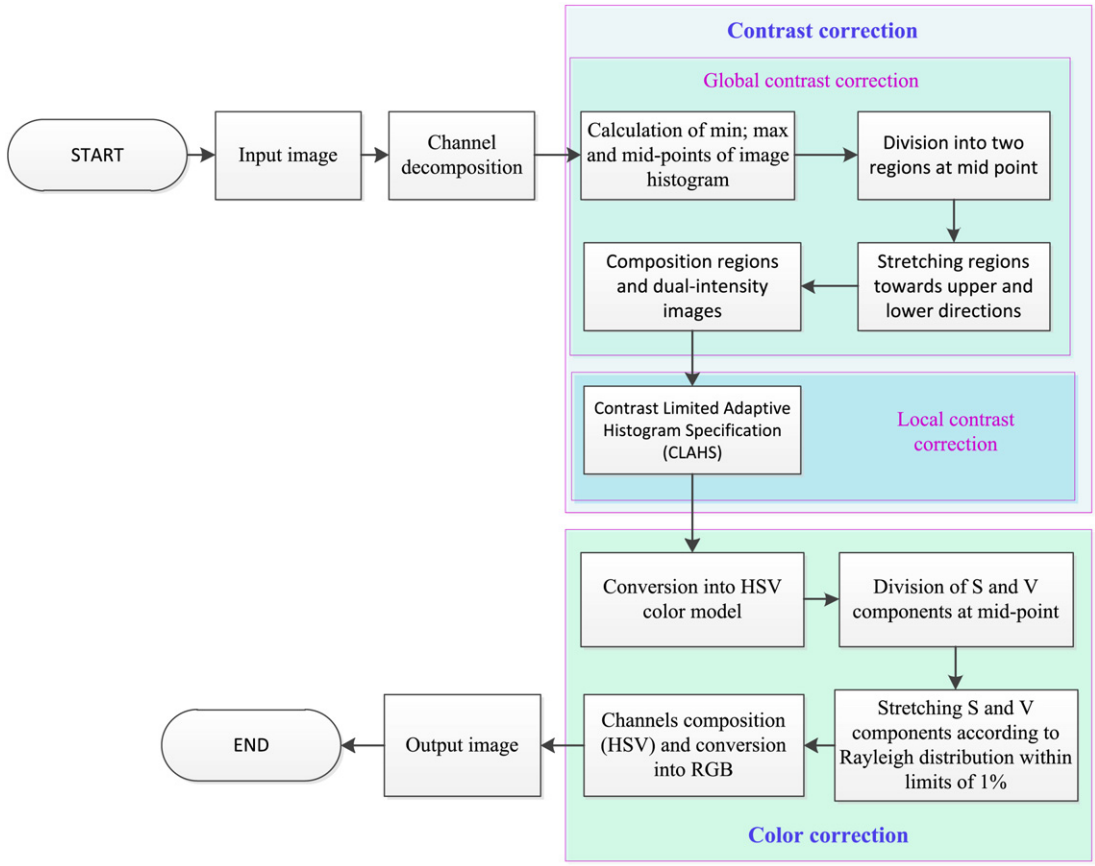

Fig. 4. Flow chart of the proposed DIRS-CLAHS technique.

4.2. Global contrast correction stage

4.2.1. Calculation of minimum, maximum, and mid-points of the histogram

At the beginning of the process, the image is decomposed into its respective channels, namely, red, green, and blue. The minimum intensity point i_{min} refers to the lowest intensity value in the image histogram, whereas the maximum intensity value i_{max} corresponds to the highest intensity value in the image histogram. The midpoint is calculated by adding the maximum and minimum intensity values and dividing the result by two; the midpoint of the intensity level i_{mid} of a histogram is thus computed using Formula (1):

$$i_{mid} = \frac{i_{min} + i_{max}}{2} \tag{1}$$

4.2.2. Division and stretching of the histogram: histogram stretching within limits

After determining the midpoint, the image histogram is divided into two regions, namely, the lower and upper regions, as depicted in Fig. 5. Then, each region is stretched as follows:

(a) The lower region is stretched toward the high intensity level with the minimum output value o_{min} of 5% from the minimum intensity value of the dynamic range (i.e., 5% from 0).
(b) The upper region is stretched toward the low intensity level with the maximum output value o_{max} of 5% from the maximum intensity value of the dynamic range (i.e., 5% from 255).

The stretching process is applied in accordance with Formula (2). P_{in} and P_{out} are the input and output pixels, respectively. i_{min}, i_{max}, o_{min}, and o_{max} are the minimum and maximum intensity level values for the input and output images, respectively.

$$P_{out} = (P_{in} - i_{min})\left(\frac{o_{max} - o_{min}}{i_{max} - i_{min}}\right) + o_{min} \tag{2}$$

4.2.3. Composition of the region and the dual-intensity images

The division and stretching processes performed in the previous step generate two histograms for each color channel. All lower-stretched histograms are composed to produce a new image in further processing; the same procedure is performed for all upper-stretched histograms to produce another image. These dual-intensity images are then integrated by using average points. Images are integrated based on the channels of the individual images, and the identical channels of two images are mapped to each other. Subsequently, the average values between these channels are calculated.

Fig. 6 illustrates the integration of under-enhanced and over-enhanced images to produce a resultant image. The histograms are processed further through local contrast stretching.

4.3. Local contrast stretching: contrast-limited adaptive histogram specification

In the local contrast stretching stage, CLAHS is applied to the histograms of the images. This method follows the following steps:

i. Division of image into tiles;
ii. Application of the clip limit;
iii. Implementation of histogram specification with respect to Rayleigh distribution;
iv. Composition of the image tiles.

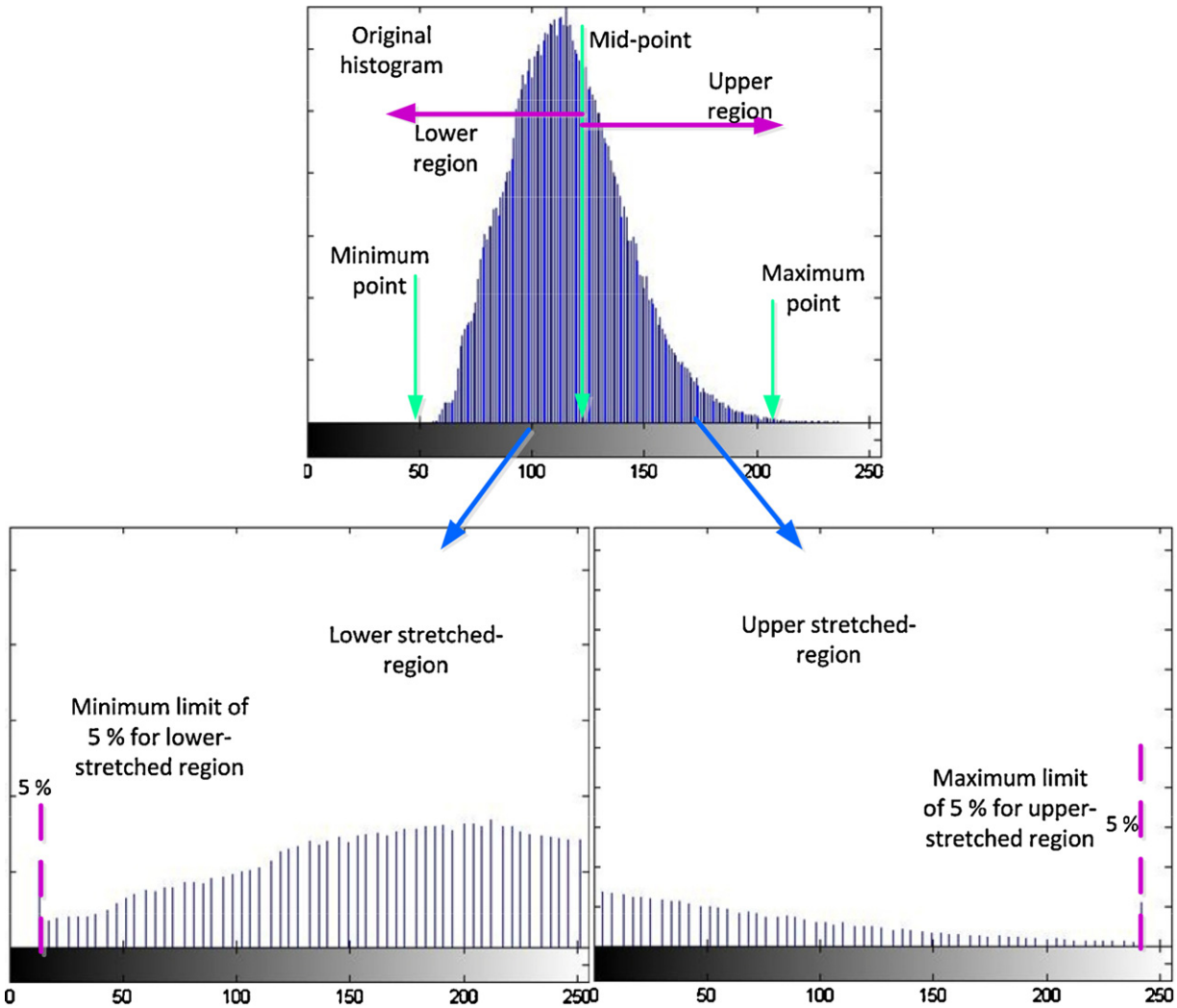

Fig. 5. Illustration of division and stretching of the original histogram to produce lower-stretched and upper-stretched regions.

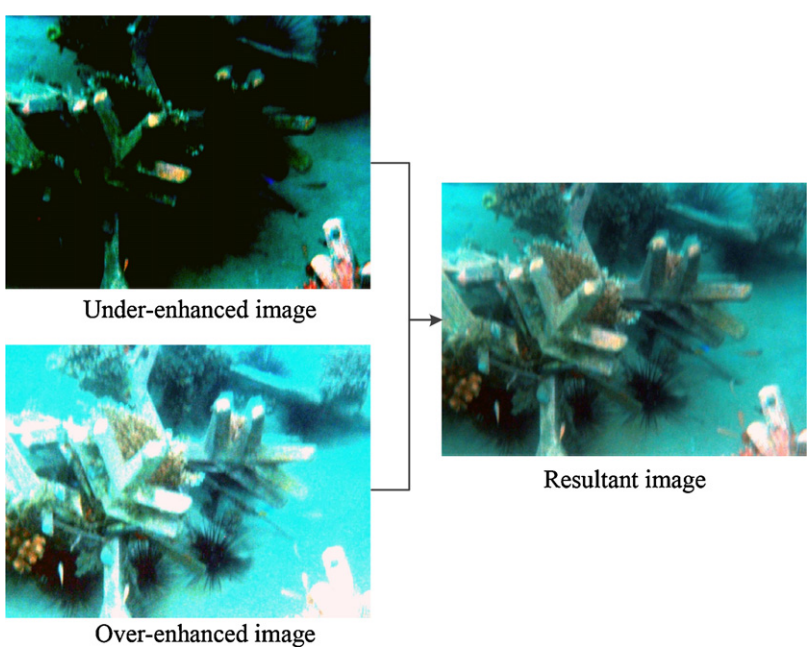

Fig. 6. Integration of under- and over-enhanced images based on average values.

Table 1
Parameter setting for the CLAHS process in the proposed method.

Parameter	Setting parameter
Number of tiles	2×2 (lowest possible tiles number)
Clip-limit	0.01 (default)
Distribution	Rayleigh
Alpha	0.4 (default)

In this processing step, the *adapthisteq* process provided in MATLAB is applied. The parameters shown in Table 1 for the proposed DIRS-CLAHS method are set in the local contrast enhancement process. The default values are determined based on suggestions derived from [7] and [19].

4.3.1. Division of the image into tiles

In the proposed DIRS-CLAHS method, the created tiles are limited to 2×2 regions, which are the smallest possible regions that can be produced through CLAHS. This value is set to avoid excessive contrast correction given that the proposed DIRS-CLAHS method already performs a preprocessing step that involves global contrast stretching. The advantage of a small number of tiles is the reduced computational processing consumption in comparison with that for the default number of 8×8 tiles. The 2×2 tiles are applied to all types of images regardless of the initial contrast level of the original image. Moreover, the histogram for each individual tile is built in consideration of the clip limit.

4.3.2. Application of the clip limit

To prevent the over-contrast effect in images, a clip limit is applied. With this limit, the number of pixels at certain intensity levels is limited to a certain value, and all the spikes in the image histogram that exceed than the clip limit are cut off. The excess pixels are equally distributed to all intensity levels, thus increasing the number of pixels at all intensity levels. This process is repeated until the number of excess pixels is sufficiently small to be ignored.

4.3.3. Remapping to Rayleigh distribution

Rayleigh distribution is ideal for underwater images [7,8,19]. This distribution refers to the bell-shaped intensity level distribution that is mostly concentrated at the middle intensity level. Intensity levels in the image histogram are mapped with respect to Rayleigh distribution after applying the clip limit. With reference to [7,8], the probability distribution function (PDF) and cumulative distribution function (CDF) of Rayleigh distribution are expressed in Formulae (3) and (4), respectively. x refers to the input pixels and α indicates the distribution parameter for Rayleigh distribution.

$$\mathrm{PDF}_{\mathrm{Rayleigh}} = \left(\frac{x}{\alpha^2}\right) e^{(-x^2/2\alpha^2)} \tag{3}$$

$$\mathrm{CDF}_{\mathrm{Rayleigh}} = 1 - e^{(-x^2/2\alpha^2)} \tag{4}$$

The image tiles are composed through bilinear interpolation [20] to limit the artificial effect between two tiles.

4.4. Application of color correction: conversion into the HSV color model

First, the image is converted into the HSV color model. This model is selected because S and V significantly influence image color and can be modified directly. Both components are also important parameters that are used to increase image clarity and visibility. As mentioned in [8,33], hue represents the image color parameter, and the modification of hues may result in inappropriate color changes in the image.

4.4.1. Division of the S and V components at the mid-point

In the HSV color model, the S and V components are stretched based on the Rayleigh distribution within a limit of 1% from the minimum and maximum intensity levels. These limits are set to prevent these components from reaching their minimum and maximum points, which can in turn lead to undersaturation and oversaturation as well as under-brightness and over-brightness. Formula (5) is applied to calculate the midpoint i_{mid} of the S and V components; i_{min} and i_{max} indicate the minimum and maximum intensity values of the S and V components, respectively.

$$i_{\mathrm{mid}} = \frac{i_{\mathrm{min}} + i_{\mathrm{max}}}{2} \tag{5}$$

Then, the histograms of the S and V components are stretched based on their midpoints.

4.4.2. Stretching of the S and V components

As mentioned previously, the S and V components are stretched according to their midpoints as follows:

(a) In the lower region, the input value is derived from the minimum intensity value i_{min} to the midpoint i_{mid}. This input region is stretched to the output histogram with reference to the new dynamic range of [1%, 99%].
(b) In the upper region, the input value is obtained from the midpoint i_{mid} to the maximum intensity value i_{max}. This input region is stretched to the output histogram according to the new dynamic range of [1%, 99%].

The S and V components are stretched based on the aforementioned steps. The stretching formula (Formula (2)) is integrated with the Rayleigh PDF (Formula (3)). The integration generates a Rayleigh-stretched formula, as expressed in Formula (6):

$$\begin{aligned}\text{Rayl.-stretched} &= \frac{[(P_{\mathrm{in}} - i_{\mathrm{min}})(o_{\mathrm{max}} - o_{\mathrm{min}}/i_{\mathrm{max}} - i_{\mathrm{min}}) + o_{\mathrm{min}}]}{\alpha^2} \\ &\quad \cdot \exp\frac{-[(P_{\mathrm{in}} - i_{\mathrm{min}})(o_{\mathrm{max}} - o_{\mathrm{min}}/i_{\mathrm{max}} - i_{\mathrm{min}}) + o_{\mathrm{min}}]^2}{2\alpha^2} \\ &= \frac{P_{\mathrm{in}}o_{\mathrm{max}} - P_{\mathrm{in}}o_{\mathrm{min}} - i_{\mathrm{min}}o_{\mathrm{max}} - o_{\mathrm{min}}i_{\mathrm{max}}}{\alpha^2(i_{\mathrm{max}} - i_{\mathrm{min}})} \\ &\quad \cdot \exp\frac{-[P_{\mathrm{in}}o_{\mathrm{max}} - P_{\mathrm{in}}o_{\mathrm{min}} - i_{\mathrm{min}}o_{\mathrm{max}} - o_{\mathrm{min}}i_{\mathrm{max}}]^2}{2\alpha^2(i_{\mathrm{max}} - i_{\mathrm{min}})^2}\end{aligned} \tag{6}$$

In the lower-stretched region, the value of i_{max} is substituted with the value of i_{mid}. By contrast, the value of i_{min} for the upper-stretched region is replaced with the value of i_{mid}.

Fig. 7 shows the H, S, and V components of the HSV color model as well as the limits applied in the proposed DIRS-CLAHS method.

4.4.3. Components composition

The lower-stretched and upper-stretched histograms of the S component in previous processes are integrated using the average value I_{avg}, as indicated in Formula (7). $I_{LS}(i, j)$ and $I_{US}(i, j)$ are the intensity values for the lower-stretched and upper-stretched histograms at positions (i, j), respectively. An identical integration process is also employed with the V component.

$$I_{avg} = \frac{I_{LS}(i, j) + I_{US}(i, j)}{2} \tag{7}$$

Then, the H, S, and V components are composed to generate an image in the HSV color model before converting it into the RGB color model. An enhanced output image is thus produced.

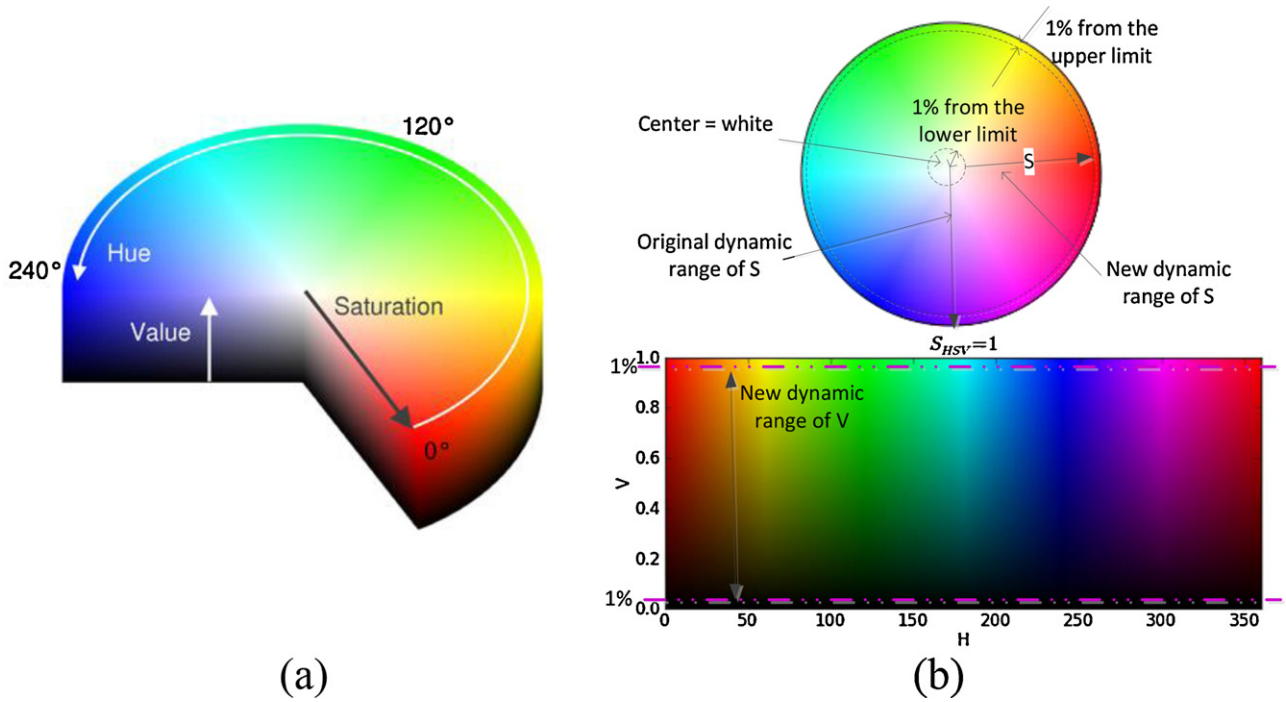

Fig. 7. HSV color model (a) illustration of the H, S, and V components of the HSV color model (b) new range of S and V components at 1% from minimum and maximum limits, which is applied in the proposed DIRS-CLAHS method.

5. Results and discussion

5.1. Qualitative and quantitative analyses

The proposed method is evaluated through qualitative observation and analysis to determine whether it can solve or reduce the problems with underwater images. A scatter diagram of an image is drawn in this paper to show the wide distribution of pixels throughout the dynamic range. A good contrast enhancement technique should distribute the image pixels throughout the entire dynamic range [11] [22].

In addition, the proposed method is assessed in terms of entropy [23], MSE [8], PSNR [8], natural image quality evaluator (NIQE) [32], and Sobel edge detection [11,28] through quantitative analysis. Entropy is defined as the corresponding gray-level states that can be adopted by individual pixels [23]. This factor represents the abundance of image information and is used for quality assessment in [23–27]. A high entropy value is preferred because it reveals that an image contains much information. In addition, an effective contrast enhancement technique should generate high PSNR and low MSE values. Sobel edge detection is also utilized in quantitative evaluation for image analysis that is evaluated based on the total number of bright pixels. A high Sobel edge detection value is preferred as it denotes the edges (objects) that can be detected in the image. MSSIM [30] is employed in the quality assessment of an image based on structural information degradation. EMEE [31] measures image enhancement or image contrast based on entropy. High MSSIM and EMEE values are preferred because they represent the structural similarity and the degree of enhancement in compared images, respectively. A low NIQE value denotes high image quality.

5.2. Performance results

In Appendix, the figures exhibiting *colorful coral, brown coral, coral stone*, and *blue fish* show four samples of the underwater images studied in the experiment, along with their respective 3D RGB color models. The quantitative results of these sample images in Appendix are presented in Table 2. Table 3 presents the average values of entropy, MSE, PSNR, Sobel edge detection, MSSIM, EMEE, and NIQE for 300 tested underwater images.

CLAHE-Mix produces a clear *colorful coral* image and improves image color, as indicated in Appendix. However, the color is unrealistic and appears unnatural, and several areas are oversaturated with red color. In addition, this method produces the second lowest entropy value after HE as well as the lowest Sobel count; as a result, images are poorly detailed. PDSCC increases contrast slightly as dark areas are retained in the resultant image. This enhancement is partial; therefore, the output and the original images do not differ significantly. MSE value is minimized as a result (153.01), and PSNR value is maximized (26.28). UCM and ICM increase image contrast to a certain extent. However, these methods fail to enhance the dark areas and generate oversaturated ones, thereby reducing image detail. The image produced by CLAHS exhibits good contrast as it is clear and the objects are effectively differentiated from the background. However, image color is diminished in the process. The HE method suffers from over-enhancement as the foreground areas brighten and the coral at the center of the image is oversaturated. The quantitative result corresponds with the observation given that the entropy value of the image produced by the HE method is the lowest among those of the other methods. In addition, the HE method generates much noise with its maximum MSE value (5838.58). This method also reports the highest Sobel count, possibly as a result of the noise produced in the process. The proposed method enhances the image effectively as the resultant image contrast is well balanced, and the areas are neither too dark nor too bright. Furthermore, the proposed method increases the image contrast at the foreground and background as both areas are significantly enhanced. The resultant image has the highest entropy value (7.761) and ranks second after PDSCC in terms of MSE (687.22) and PSNR (19.76) values. These values confirm that noise is limited in the resultant image.

Table 2
Comparison of quantitative results of the resultant images for the respective methods as shown in Appendix.

Image	Method	Quantitative analysis						
		Entropy (bit)	MSE	PSNR (dB)	Sobel count	MSSIM	EMEE	NIQE
Colorful coral	Original	7.287	–	–	42,522	–	–	–
	CLAHE-Mix	7.139	1931.75	15.27	39,392	0.747	1.701	3.108
	PDSCC	7.338	**153.01**	**26.28**	47,608	**0.986**	2.335	3.126
	UCM	7.612	1090.44	17.75	72,063	0.903	4.973	2.976
	ICM	7.610	959.90	18.31	69,176	0.921	4.963	3.036
	CLAHS	7.627	2835.62	13.60	87,838	0.686	2.923	3.226
	HE	5.985	5838.58	10.47	**92,918**	0.611	4.301	**2.903**
	DIRS-CLAHS	**7.761**	687.22	19.76	79,216	0.894	**5.338**	3.049
Brown coral	Original	7.492	–	–	16,504	–	–	–
	CLAHE-Mix	7.179	562.22	20.63	24,879	0.954	1.199	3.129
	PDSCC	7.343	**369.09**	**22.46**	18,882	**0.986**	1.251	2.979
	UCM	**7.683**	3411.33	12.80	37,712	0.772	4.815	2.950
	ICM	7.618	1812.00	15.55	35,816	0.804	8.251	2.970
	CLAHS	7.420	1481.68	16.42	**41,090**	0.799	2.937	3.049
	HE	5.982	3534.60	12.62	40,995	0.654	6.486	**2.780**
	DIRS-CLAHS	7.673	509.54	21.06	40,962	0.949	**9.982**	3.014
Coral stone	Original	7.024	–	–	10,203	–	–	–
	CLAHE-Mix	7.194	348.77	22.71	13,506	0.966	3.795	4.012
	PDSCC	6.843	1371.79	16.76	16,051	0.941	0.460	3.915
	UCM	7.374	7259.26	9.52	35,256	0.530	6.615	3.934
	ICM	**7.600**	4748.93	11.36	34,982	0.539	6.792	3.914
	CLAHS	7.124	1142.48	17.55	25,983	0.809	3.237	**3.882**
	HE	5.907	6826.12	9.79	**38,671**	0.537	10.710	4.173
	DIRS-CLAHS	7.465	**318.78**	**23.11**	32,611	**0.979**	**22.269**	4.110
Blue fish	Original	7.388	–	–	14,173	–	–	–
	CLAHE-Mix	7.095	438.47	21.42	7095	0.967	1.446	2.571
	PDSCC	7.117	652.08	19.99	17,127	**0.976**	1.031	2.568
	UCM	7.579	5528.92	10.70	40,198	0.622	8.961	2.520
	ICM	7.564	4119.97	11.98	38,813	0.626	**9.635**	2.639
	CLAHS	7.374	1396.80	16.68	37,377	0.819	2.736	**2.383**
	HE	5.955	4320.71	11.78	**44,306**	0.624	5.182	2.424
	DIRS-CLAHS	**7.668**	**412.12**	**21.98**	41,027	0.962	7.125	2.470

Note: The values in bold typeface represent the best result obtained in the comparison.

CLAHE-Mix increases contrast slightly in the *brown coral* image while retaining the blue-green illumination effect. The colors of the objects remain influenced by the blue illumination in water, and the blue-green illumination effect of the PDSCC method can still be observed. Although PDSCC generates the highest PSNR (22.46) values among the studied methods, image contrast did not increase significantly. The resultant images produced by CLAHE-Mix and PDSCC did not vary considerably because the effect of blue-green illumination is retained. The UCM increases the image contrast, and the blue-green illumination effect is mostly eliminated. As such, the UCM ranks first in terms of entropy with a value of 7.683. However, the resultant image from this method is yellowish as a result of excessive red color. The ICM improves image contrast as the objects are clear; however, this method enhances contrast at the foreground more than at the background. Moreover, the foreground is excessively enhanced as the stone becomes oversaturated. Meanwhile, the image produced by CLAHS exhibits good contrast; it is clear, and the objects can be distinguished from the background. Thus, CLAHS reports the highest Sobel count at 41,090. However, the resultant image displays low image color, and the colors are not widely distributed throughout the RGB color model. S and V values are also low. The HE method produces excessive red color and renders the resultant image reddish. As the best method among the examined approaches, the proposed DIRS-CLAHS method exhibits improved image contrast and color; this method significantly increases image contrast as well given that the objects in the image are well distinguished from the background.

Table 3
Average quantitative values of 300 underwater images in comparison with state-of-the-art methods.

Method	Quantitative analysis						
	Entropy (bit)	MSE	PSNR (dB)	Sobel count	MSSIM	EMEE	NIQE
Original	7.287	–	–	42,522	–	–	–
CLAHE-Mix	7.139	1931.75	15.27	39,392	0.895	1.522	3.627
PDSCC	7.338	**153.01**	**26.28**	47,608	**0.926**	0.947	3.982
UCM	7.612	1090.44	17.75	72,063	0.648	3.538	4.037
ICM	7.610	959.90	18.31	69,176	0.695	4.455	3.915
CLAHS	7.627	2835.62	13.60	87,838	0.810	1.799	3.645
HE	5.985	5838.58	10.47	**92,918**	0.609	5.287	**2.559**
DIRS-CLAHS	**7.761**	687.22	19.76	79,216	0.910	**9.585**	3.931

Note: The values in bold typeface represent the best result obtained in the comparison.

Effects similar to the first two image samples are observed on the resultant *coral stone* and *blue fish* images. The quantitative evaluation results presented in Table 2 support the visual observation of the images. Overall, the proposed method successfully enhances image contrast and improves image details; the average quantitative results obtained from 300 underwater image samples (Table 3) support the capability and performance of the proposed method because DIRS-CLAHS has the highest entropy value of 7.761 in comparison with the other state-of-the-art methods. In terms of MSE and PSNR values, the proposed method is ranked second after PDSCC with reported values of 687.22 and 19.76, respectively. The qualitative results indicate that PDSCC fails to enhance the quality of underwater images as this method produces images with low contrast and details. With respect to MSSIM value, the proposed method ranks second (0.926) after PDSCC. The proposed method also generates the highest EMEE value of 9.585. All these findings favor the proposed DIRS-CLAHS method as the best contrast enhancement method for underwater images over the other tested state-of-the-art methods.

6. Conclusion

The proposed DIRS-CLAHS method integrates contrast correction and color correction methods. In the contrast correction step, this method incorporates the processes of global and local contrast stretching to enhance image contrast. The color correction step improves the image color performance determined in previous steps. The qualitative and quantitative results suggest that the proposed technique successfully improves image contrast, reduces image noise, and weakens the effect of under-enhanced and over-enhanced areas. In addition, the DIRS-CLAHS method outperforms current state-of-the-art methods in terms of image color. Undersaturation and oversaturation effects are also significantly weakened in the images produced with the proposed method. Based on its promising performance, the proposed DIRS-CLAHS method may improve on and further research in the area of underwater image processing. Nevertheless, all the tested methods exhibit color image distortion when an image has a low color percentage (less than 1%), particularly with respect to the red color channel. This limitation will be considered in future works on the proposed method.

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Appendix A. Appendix

Figs. A1 and A2

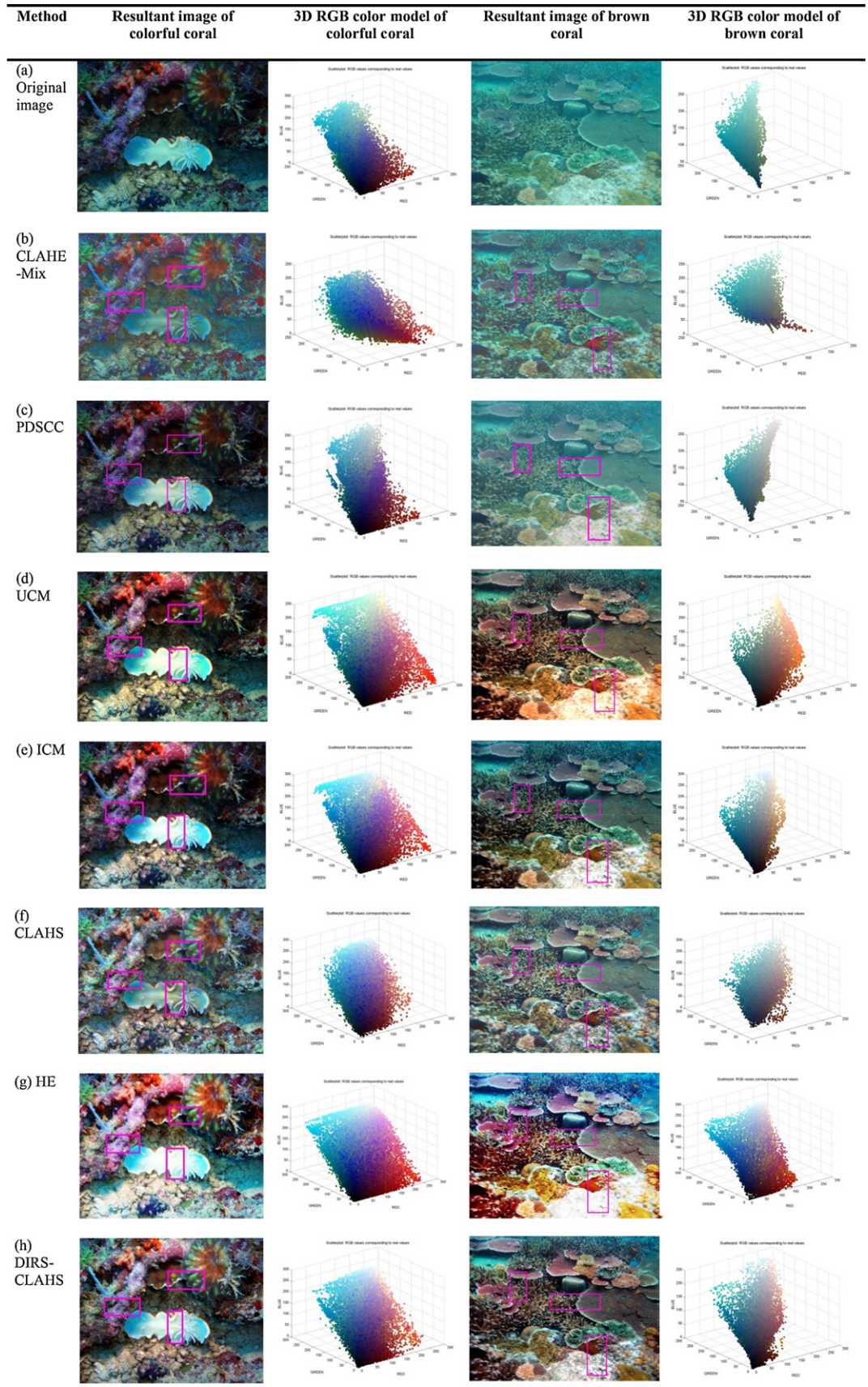

Fig. A1. Images of *colorful coral* and *brown coral* tested with different methods with their 3D RGB color model.

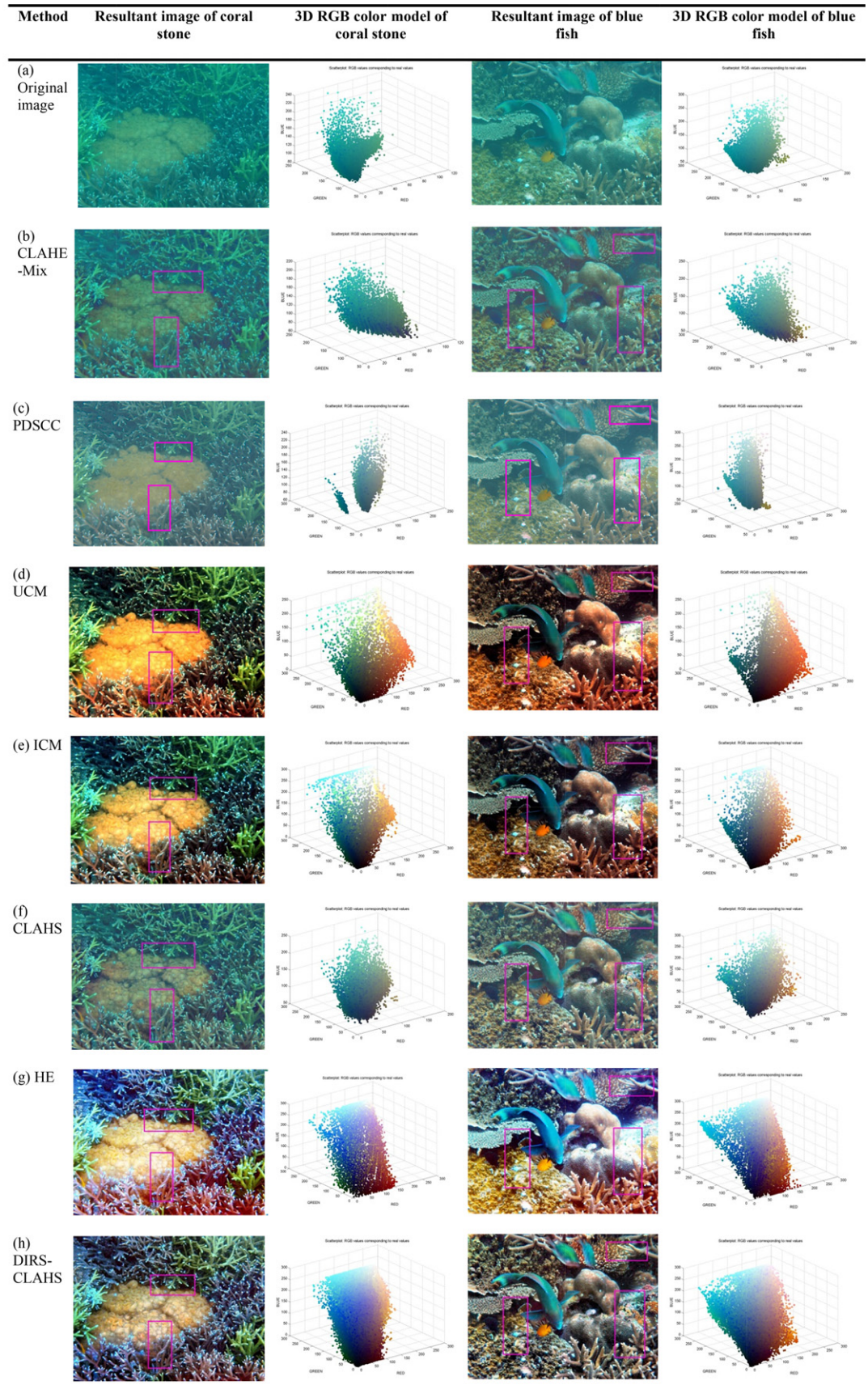

Fig. A2. Images of *coral stone* and *blue fish* tested with different methods with their 3D RGB color model.

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